Elektronik II

Feedback: General Structure, Voltage and Current Amplifiers

Original notes: Ders11.pdf

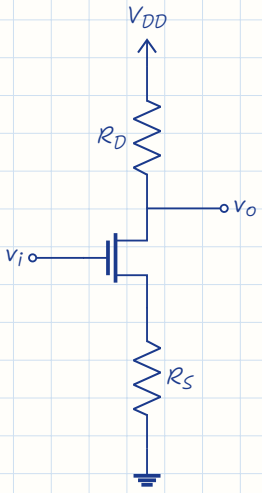
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Not: Orijinal dosyadaki basılı sayfalar (s. 2–5) dijitalleştirilmedi; yalnızca el yazısı sayfalar aktarıldı.

orig. p. 1

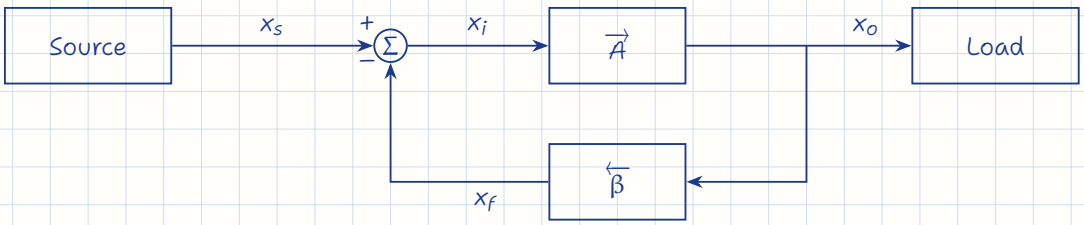
Feedback



Common source amplifier

$$\frac{v_o}{v_i} \approx \frac{R_D}{R_S}$$

The general feedback structure

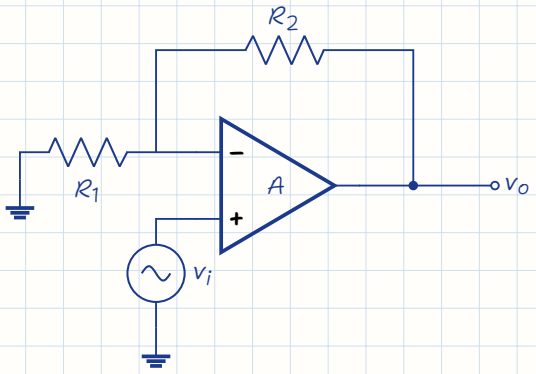


A = open loop gain
β = feedback factor
x_i = Input signal
x_s = signal source
x_o = output signal
x_f = sampled output by fbn (feedback network)

$$\left(\frac{x_o}{x_s}\right)_{CL} = ?$$

$$x_o = A x_i$$
$$x_f = \beta \cdot x_o$$
$$x_i = x_s - x_f$$
$$x_i = x_s - \beta x_o$$
$$x_i = x_s - A\beta \cdot x_i$$
$$x_s = x_i (1 + A\beta)$$

$$\left(\frac{x_o}{x_s}\right) = \frac{A x_i}{x_i (1 + A\beta)} = \frac{A}{1 + A\beta}$$



$$\frac{v_o}{v_i} = \left(1 + \frac{R_2}{R_1}\right) = \frac{1}{\beta}$$

orig. p. 6

1- Voltage Amplifiers (Shunt (o) – Series (I))

Figure: Block diagram of the shunt-series voltage amplifier: source V_s with series R_s feeds the "Basic Amplifier" block; the amplifier output is connected in parallel with the load R_L and the output terminals $(+V_o-)$; the "Feedback" block has its input $(+V_o-)$ connected across the output (shunt), and its output voltage V_f (+ top, – bottom) is in series with the input loop (between V_s/R_s and the amplifier input). Orange arrows mark R_{if} (looking into the source loop), R_{in} (looking into the amplifier input), R_{out} (looking into the amplifier output, at R_L) and R_{of} (looking into the output terminals).
(in the original notes, p. 6)

Two port network equivalent

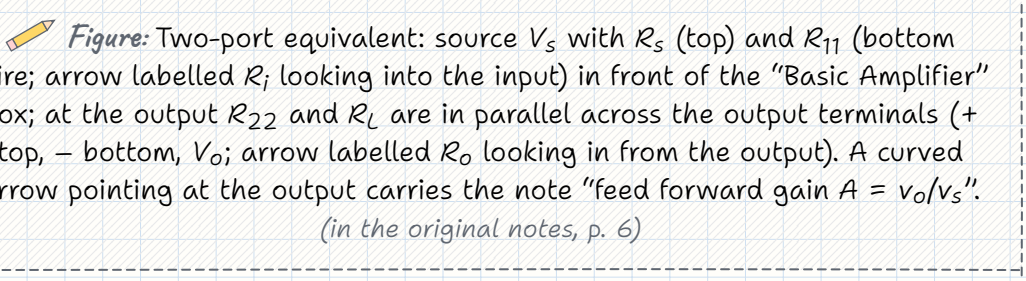


Figure: Two-port equivalent: source V_s with R_s (top) and R_{11} (bottom wire; arrow labelled R_i looking into the input) in front of the "Basic Amplifier" box; at the output R_{22} and R_L are in parallel across the output terminals (+ top, − bottom, V_o; arrow labelled R_o looking in from the output). A curved arrow pointing at the output carries the note "feed forward gain $A = v_o/v_s$".

(in the original notes, p. 6)

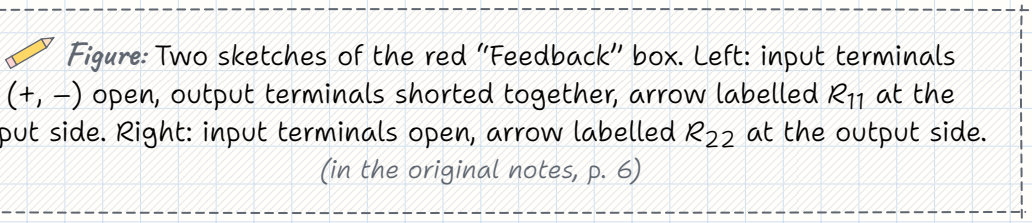


Figure: Two sketches of the red "Feedback" box. Left: input terminals (+, −) open, output terminals shorted together, arrow labelled R_{11} at the input side. Right: input terminals open, arrow labelled R_{22} at the output side.

(in the original notes, p. 6)

To find β

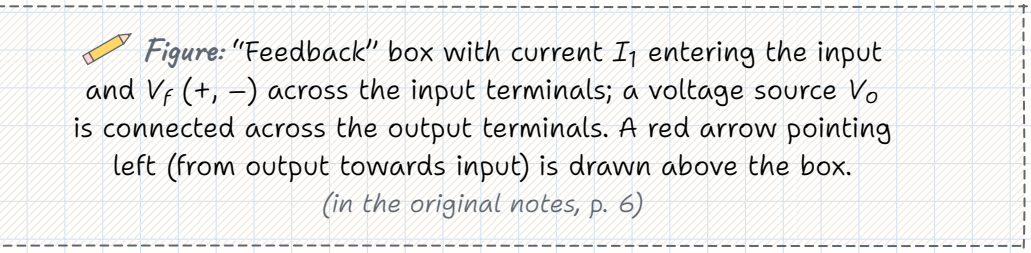


Figure: "Feedback" box with current I_1 entering the input and V_f (+, −) across the input terminals; a voltage source V_o is connected across the output terminals. A red arrow pointing left (from output towards input) is drawn above the box.

(in the original notes, p. 6)

$$\beta = \left.\frac{V_f}{V_o}\right|_{I_1=0}$$

orig. p. 7

Complete model

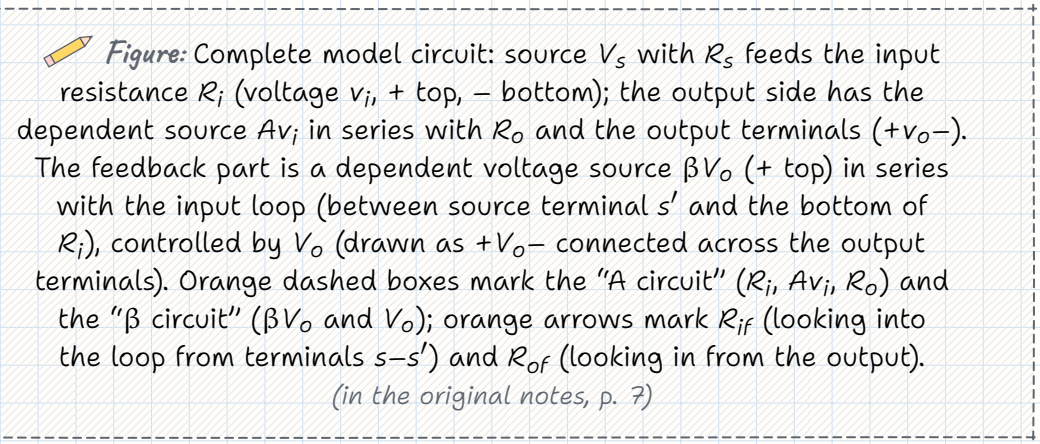


Figure: Complete model circuit: source V_s with R_s feeds the input resistance R_i (voltage v_i, + top, − bottom); the output side has the dependent source Av_i in series with R_o and the output terminals ($+v_o-$). The feedback part is a dependent voltage source βV_o (+ top) in series with the input loop (between source terminal s' and the bottom of R_i), controlled by V_o (drawn as $+V_o-$ connected across the output terminals). Orange dashed boxes mark the "A circuit" (R_i, Av_i, R_o) and the "β circuit" (βV_o and V_o); orange arrows mark R_{if} (looking into the loop from terminals s–s') and R_{of} (looking in from the output).

(in the original notes, p. 7)

Figure: Two reduced circuits. Left: source V_s (+ top, − bottom; terminals s and s') with $\mathcal{R}_s$ (struck out in red) in series with $\mathcal{R}_{if}$. Right: dependent source $A_f V_s$ in series with $\mathcal{R}_{of}$ (top wire, struck out in red); output $+v_o-$ across a resistor labelled $\mathcal{R}_L$ (struck out in red) with the red label $\mathcal{R}_{of}$ next to it.
(in the original notes, p. 7)

$$A_f = \frac{A}{1 + A\beta}$$
$$\mathcal{R}_{if} = (1 + A\beta)\,\mathcal{R}_i$$
$$\mathcal{R}_{of} = \frac{\mathcal{R}_o}{1 + A\beta}$$

orig. p. 8

2- Current Amplifiers (Series (o) – Shunt (I))

$$A = \frac{I_o}{I_i}$$

Figure: Block diagram of the series-shunt current amplifier: current source I_s with parallel $\mathcal{R}_s$ feeds the "Basic current Amplifier" block; the output current flows through the load $\mathcal{R}_L$. The feedback network ("FBN") takes the output current I_o in series with the load path (orange oval and orange note "Series") and returns its output to the input node in parallel with $I_s/\mathcal{R}_s$ (orange oval and orange note "Shunt").
(in the original notes, p. 8)

Figure: Two-port equivalent: I_s with $\mathcal{R}_s$ and $\mathcal{R}_{11}$ in parallel (arrow labelled $\mathcal{R}_i$ looking into the input) feeds the "Basic Circuit Amplifier" box; at the output $\mathcal{R}_{22}$ is in series with the load $\mathcal{R}_L$, current I_o (arrow) in the output loop, and an arrow labelled $\mathcal{R}_o$ looking in at the output terminals.
(in the original notes, p. 8)

Figure: Two sketches of the "FBN" box. Left: arrow labelled $\mathcal{R}_{11}$ at the input side, output terminals open. Right: input terminals shorted together, arrow labelled $\mathcal{R}_{22}$ at the output side.
(in the original notes, p. 8)

Figure: "FBN" box with input current I_f entering and voltage v_1 (+, −) across the input terminals; a current source I_o is connected at the output.
(in the original notes, p. 8)

$$\beta = \left.\frac{I_f}{I_o}\right|_{v_1=0}$$

Figure: Complete model: current source I_s (terminals s and s') feeds $\mathcal{R}_i$ (voltage v_i, + top, − bottom; input current I_i). Output side: dependent current source AI_i in parallel with $\mathcal{R}_o$, ending at the output terminals. Feedback part: dependent current source βI_o connected across the input nodes (parallel with $\mathcal{R}_i$), with the output current I_o (arrow) flowing in a series path at the output terminals. Orange dashed boxes: "A circuit" ($\mathcal{R}_i$, AI_i, $\mathcal{R}_o$) and "β circuit" (βI_o and I_o); orange arrows mark $\mathcal{R}_{if}$ (looking into s–s') and $\mathcal{R}_{of}$ (looking in from the output).
(in the original notes, p. 8)

orig. p. 9

Figure: Two small circuits: left, terminals s and s' with the current I_s (arrow) entering $\mathcal{R}_{if}$; right, dependent current source $A_f I_s$ in parallel with $\mathcal{R}_{of}$, with the output current I_o (arrow).
(in the original notes, p. 9)

$$A_f = \frac{A}{1 + A\beta}$$

$$\mathcal{R}_{if} = \frac{\mathcal{R}_i}{1 + A\beta}$$

$$\mathcal{R}_{of} = \mathcal{R}_o\,(1 + A\beta)$$

$$1 + \frac{\mathcal{R}_2}{\mathcal{R}_1} = 1001$$

Soru

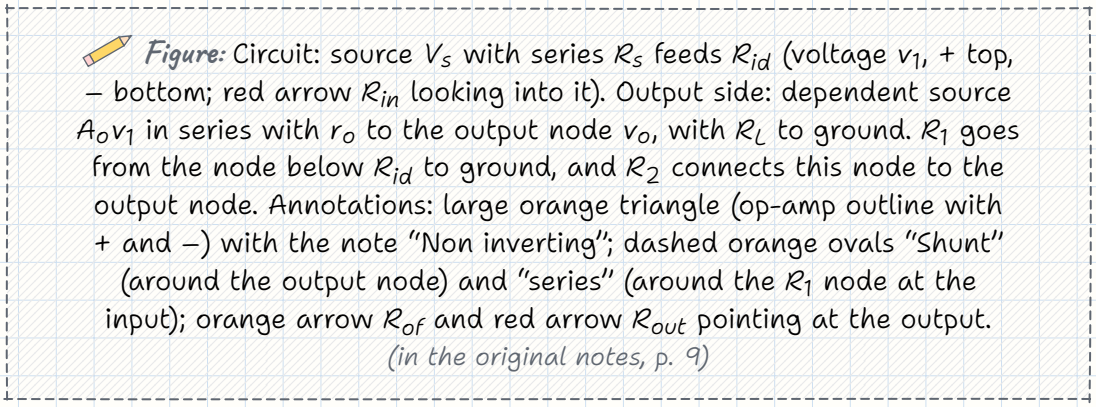


Figure: Circuit: source V_s with series R_s feeds R_{id} (voltage v_1, + top, − bottom; red arrow R_{in} looking into it). Output side: dependent source $A_o v_1$ in series with r_o to the output node v_o, with R_L to ground. R_1 goes from the node below R_{id} to ground, and R_2 connects this node to the output node. Annotations: large orange triangle (op-amp outline with + and −) with the note "Non inverting"; dashed orange ovals "Shunt" (around the output node) and "series" (around the R_1 node at the input); orange arrow R_{of} and red arrow R_{out} pointing at the output.
(in the original notes, p. 9)

$$R_{in} = R_{if} - R_s \qquad R_{out} = \frac{1}{1/R_{of} - 1/R_L}$$

Find expressions for
A, β, A_f, R_i, R_{if}, R_{of}, and R_o
($A_o = 10^4$, $R_{id} = 100\,\text{k}\Omega$, $r_o = 1\,\text{k}\Omega$, $R_L = 2\,\text{k}\Omega$, $R_1 = 1\,\text{k}\Omega$, $R_2 = 1\,\text{M}\Omega$ and $R_s = 10\,\text{k}\Omega$)
Feedback-type : Shunt − Series

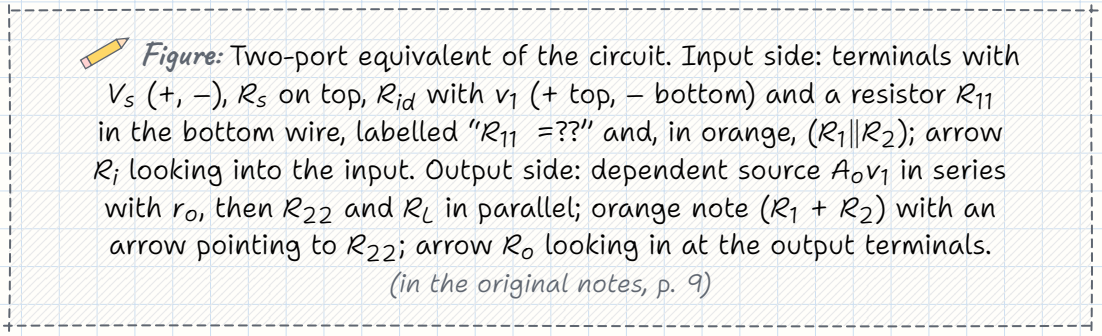


Figure: Two-port equivalent of the circuit. Input side: terminals with V_s (+, −), R_s on top, R_{id} with v_1 (+ top, − bottom) and a resistor R_{11} in the bottom wire, labelled "R_{11} =??" and, in orange, $(R_1 \| R_2)$; arrow R_i looking into the input. Output side: dependent source $A_o v_1$ in series with r_o, then R_{22} and R_L in parallel; orange note $(R_1 + R_2)$ with an arrow pointing to R_{22}; arrow R_o looking in at the output terminals.
(in the original notes, p. 9)

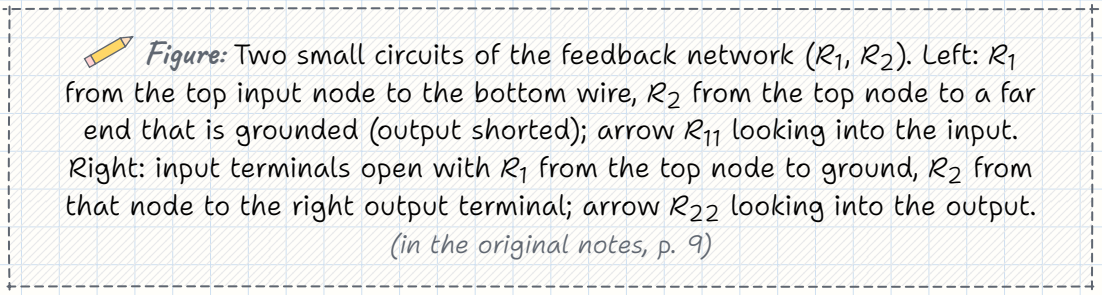


Figure: Two small circuits of the feedback network (R_1, R_2). Left: R_1 from the top input node to the bottom wire, R_2 from the top node to a far end that is grounded (output shorted); arrow R_{11} looking into the input. Right: input terminals open with R_1 from the top node to ground, R_2 from that node to the right output terminal; arrow R_{22} looking into the output.
(in the original notes, p. 9)

$$R_{11} = R_1 \parallel R_2 \qquad R_{22} = R_1 + R_2$$

orig. p. 10

Figure: Circuit: $\mathcal{R}_1$ in the left vertical branch with v_f (+ top, − bottom) across it; in the top wire a resistor (labelled $\mathcal{R}_1$ in the note, presumably $\mathcal{R}_2$) in series with a voltage source v_o on the right.
(in the original notes, p. 10)

$$\frac{v_f}{v_o} = \frac{\mathcal{R}_1}{\mathcal{R}_1 + \mathcal{R}_2} = \beta$$

$$\mathcal{R}_i = \mathcal{R}_s + \mathcal{R}_{id} + \mathcal{R}_{11}$$
$$\mathcal{R}_o = r_o \parallel \mathcal{R}_{22} \parallel \mathcal{R}_L$$

$$A = \underbrace{\frac{\mathcal{R}_{id}}{\mathcal{R}_i}}_{\frac{v_o}{v_s} = \frac{v_1}{v_s}} \cdot \underbrace{\frac{\mathcal{R}_{22} \parallel \mathcal{R}_L}{r_o + \mathcal{R}_{22} \parallel \mathcal{R}_L}}_{\frac{v_o}{v_1} \text{ katkıları}} \cdot A_o$$

$$A_f = \frac{A}{1 + A\beta}$$

$$A = \frac{100\,\text{k}\Omega}{111\,\text{k}\Omega} \cdot \frac{2\,\text{k}\Omega}{3\,\text{k}\Omega} \cdot 10^4 \approx 6000\ [\text{V/V}]$$

$\mathcal{R}_i = 111\text{k}$
$\mathcal{R}_o = 0{,}60\text{k}$

$$\beta = \frac{\mathcal{R}_1}{\mathcal{R}_1 + \mathcal{R}_2} = 10^{-3}\ [\text{V/V}]$$
$$1 + A\beta = 1 + 6000 \cdot 10^{-3} = 7$$
$$A_f = \frac{6000}{7} = 857\ [\text{V/V}]$$

$$\mathcal{R}_{if} = (1 + A\beta) \overset{111\text{k}}{\mathcal{R}_i} = 777\,\text{k}\Omega$$
$$\mathcal{R}_{of} = \frac{\mathcal{R}_o}{(1 + A\beta)} \approx 93\,\Omega$$